

SIMATS ENGINEERING

ASSESSMENT TOOL 2

Annexure A - Concept Mapping

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Register Number	192473028
Course Name	Fundamentals of Data Science
Course Code	DSA0402
Course Outcome Covered	CO1: Analyse the role of data science, facets of data, and the big data ecosystem for informed decision-making. (BL4)
Number of Questions	05
Total Marks	100

Code of Conduct

I V.Shiva (Reg No: 192472256) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

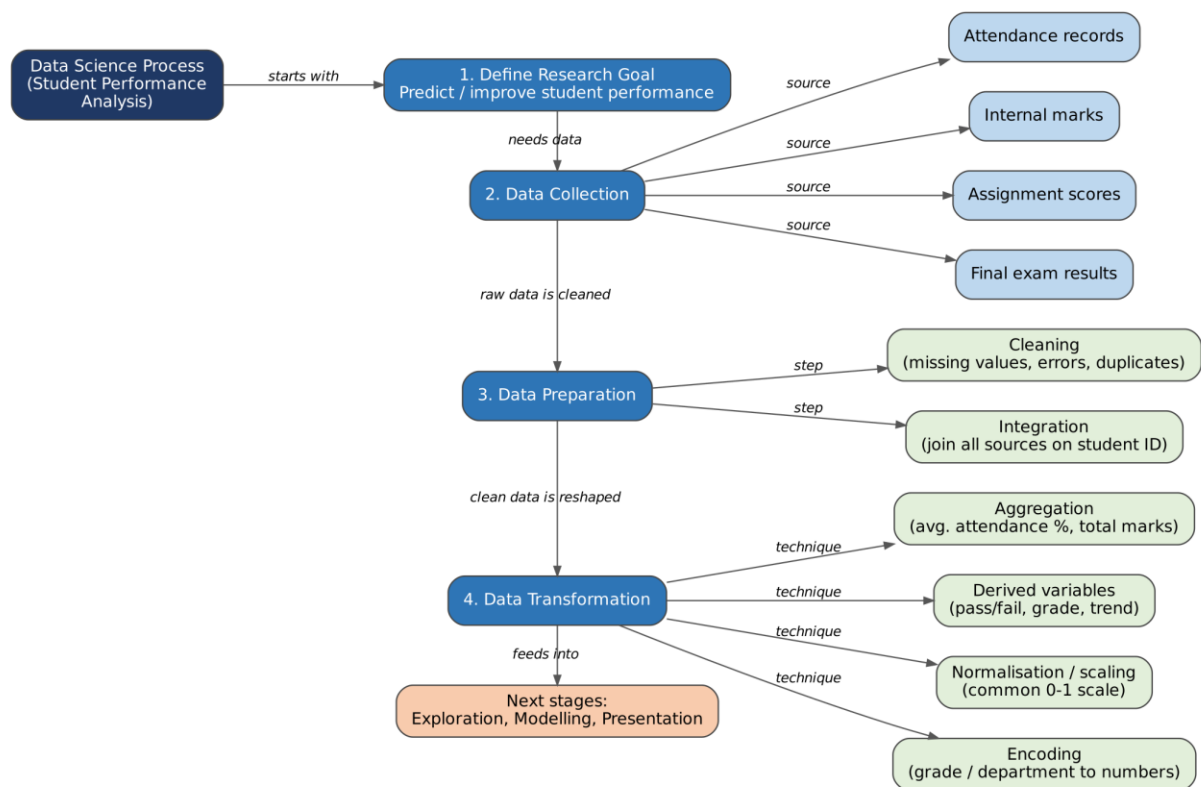
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ____Shiva_____

Question 1

A college wants to analyze student performance using attendance records, internal marks, assignment scores, and final exam results. Draw a concept map showing the Data Science process from data collection to data transformation.

Concept Map



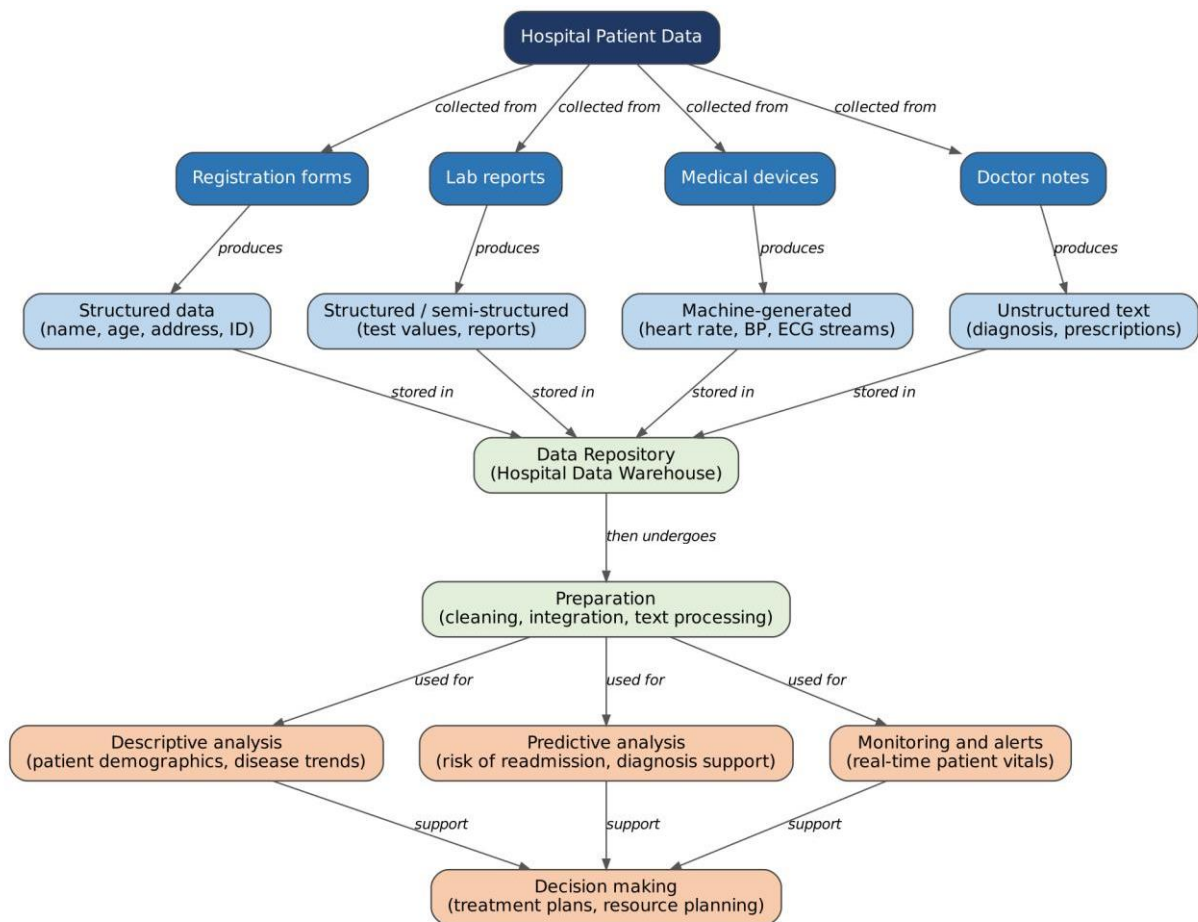
Explanation

- **Define goal:** the process starts by stating what to find out, e.g. predict or improve student performance.
- **Collect:** attendance, internal marks, assignment scores and final exam results are gathered as raw data.
- **Prepare:** cleaning removes missing values, errors and duplicates; integration joins all sources using student ID.
- **Transform:** aggregation, derived variables, scaling and encoding turn clean data into model-ready form.

Question 2

A hospital collects patient details from registration forms, lab reports, medical devices, and doctor notes. Construct a concept map showing the different data collection sources and how they are used for analysis.

Concept Map



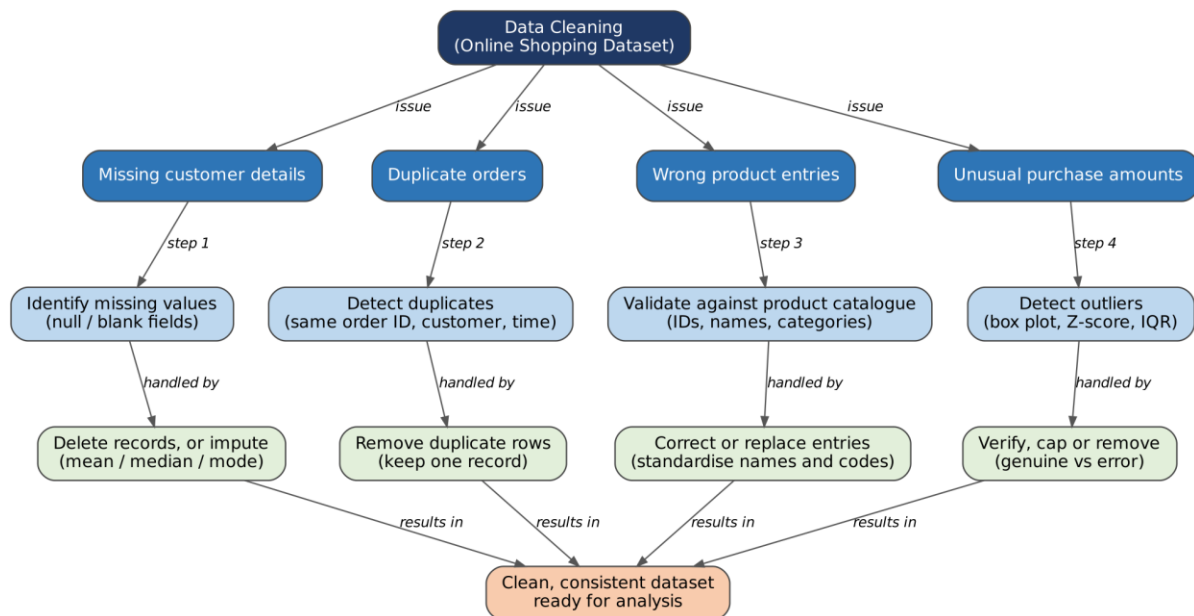
Explanation

- **Sources and types:** forms give structured data, lab reports give structured or semi-structured data, devices give machine-generated streams, doctor notes give unstructured text.
- **Storage:** all data is stored together in a central repository such as a data warehouse.
- **Use:** after preparation it supports descriptive analysis, predictive analysis and real-time monitoring, which in turn support medical and management decisions.

Question 3

An online shopping company finds missing customer details, duplicate orders, wrong product entries, and unusual purchase amounts in its dataset. Draw a concept map showing the data cleaning steps required before analysis.

Concept Map



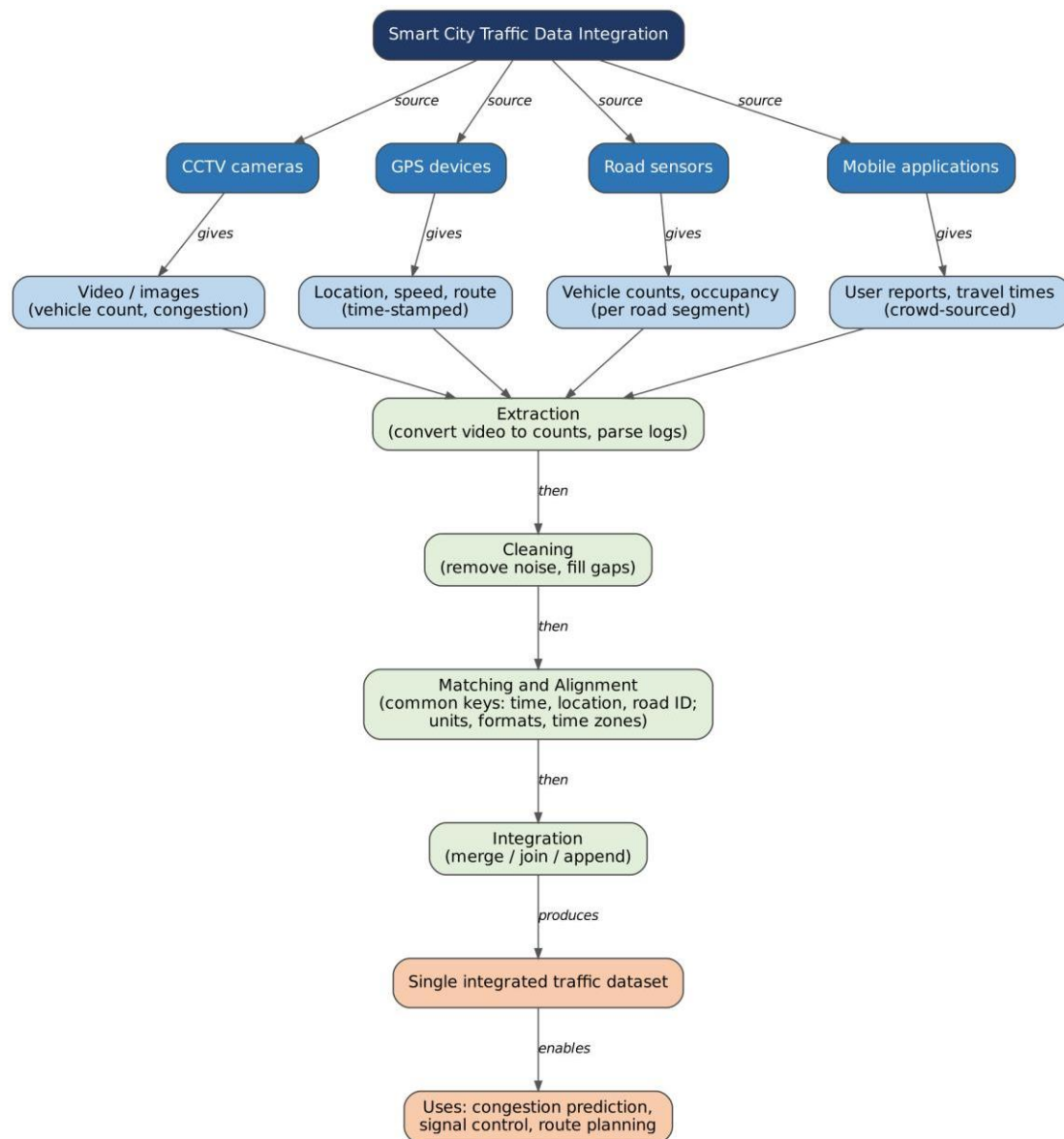
Explanation

- **Missing details:** identify null or blank fields, then delete the records or impute using mean, median or mode.
- **Duplicates:** detect repeated orders by order ID, customer and time, and keep one record.
- **Wrong entries:** validate against the product catalogue and standardise names and codes.
- **Unusual amounts:** detect outliers using box plot, Z-score or IQR, then verify and cap or remove genuine errors.

Question 4

A smart city project collects traffic data from CCTV cameras, GPS devices, sensors, and mobile applications. Prepare a concept map showing how data from multiple sources can be integrated into a single dataset.

Concept Map



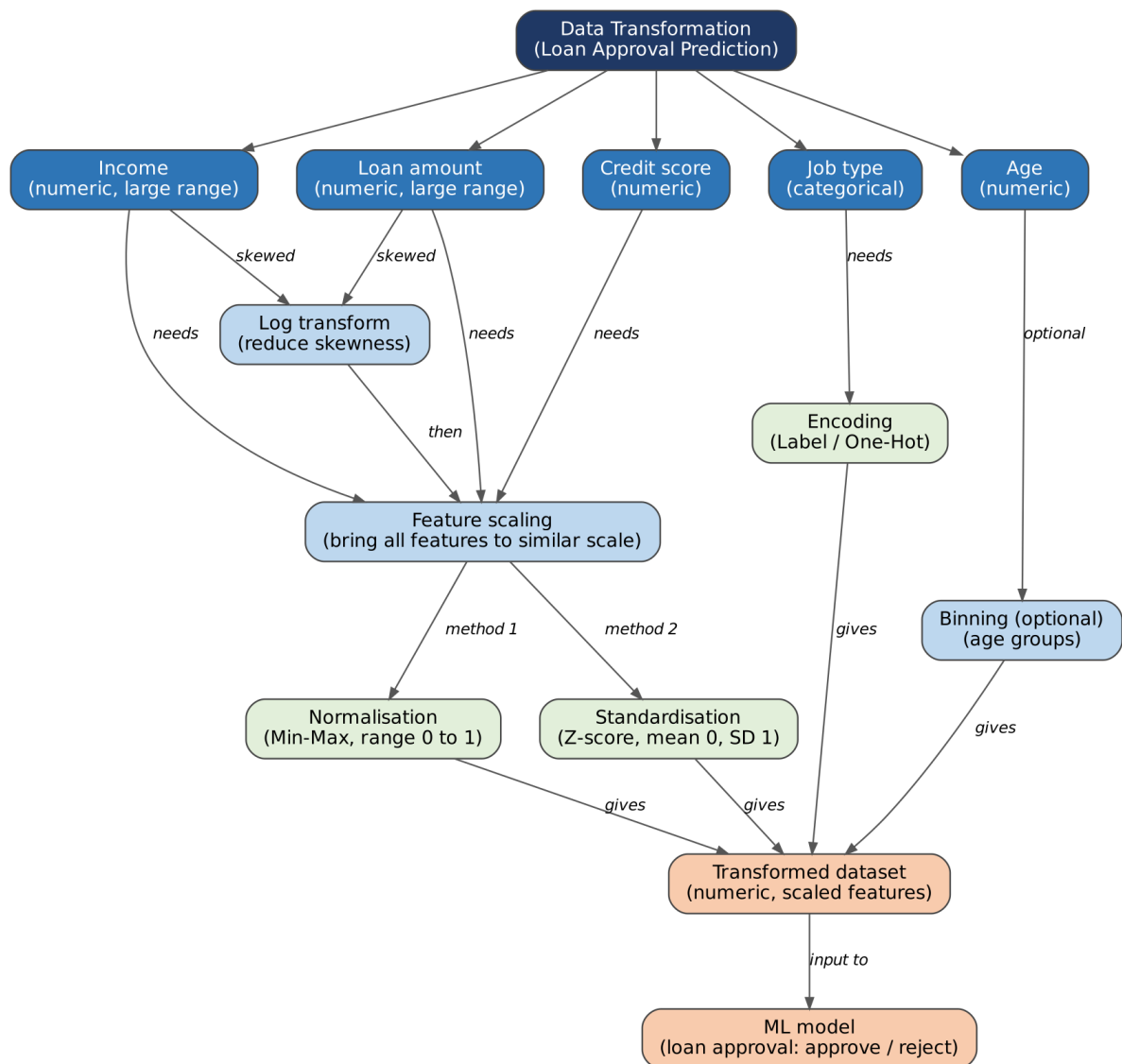
Explanation

- **Different formats:** video, location logs, sensor counts and user reports must first be converted into comparable records.
- **Common keys:** time, location and road ID are used to match records; units, formats and time zones are aligned.
- **Integration:** the cleaned and aligned data is merged, joined or appended into one traffic dataset used for congestion prediction, signal control and route planning.

Question 5

A bank wants to prepare customer data for loan approval prediction. The dataset contains income, age, job type, loan amount, and credit score. Draw a concept map showing suitable data transformation techniques such as encoding, normalization, and feature scaling.

Concept Map



Explanation

- **Encoding:** job type is categorical, so Label or One-Hot encoding converts it to numbers.
- **Feature scaling:** income, loan amount and credit score have very different ranges, so they are scaled using normalisation (Min-Max, 0 to 1) or standardisation (Z-score).
- **Other:** log transform reduces skew in income and loan amount; age may optionally be binned into groups.
- **Result:** a numeric, scaled dataset that is the input to the loan approval model.